ECR & ECS with Fargate

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[참조가이드]

ECR 참조

https://docs.aws.amazon.com/ko_kr/AmazonECR/latest/userguide/what-is-ecr.html

ECS 참조

https://docs.aws.amazon.com/AmazonECS/latest/developerguide/Welcome.html

API-Guide

https://docs.aws.amazon.com/ko_kr/AmazonECS/latest/APIReference/Welcome.html

ECS-CLI

https://docs.aws.amazon.com/cli/latest/reference/ecs/index.html

[ECS 개념이해]

https://docs.aws.amazon.com/AmazonECS/latest/developerguide/create-container-image.html

[ECS 설정인터페이스]

- Amazon ECS 콘솔
- AWS Command Line Interface (AWS CLI)
- AWS SDK
- AWS Copilot

[ECS-Task 정의]

▶ Why? "ECS +Fargate"

AWS 환경에서 신규 컨테이너 기반 서비스를 구축할 경우, Amazon ECS 와 AWS Fargate 를 조합한 아키텍처가 가장 권장되는 운영 방식 중 하나로 평가된다.

ECS + Fargate 구성은 서버 및 노드 관리 부담 없이 컨테이너를 실행할 수 있으며, Auto Scaling, IAM, ALB, CloudWatch, Aurora 등 AWS 핵심 서비스와 높은 수준으로 통합된다.

특히 Kubernetes(EKS) 대비 운영 복잡도가 낮고 구축 속도가 빠르기 때문에, 일반적인 웹 서비스·API 서비스·마이크로서비스 환경에서는 AWS 의 대표적인 실무형 컨테이너 아키텍처로 널리 사용된다.

▶ Task 제약사항 for AWS Fargate

- 운영체제 : Amazon Linux 2, Windows Server 2019&2022 --> 실무에서는 대부분 Linux Fargate 사용
- CPU 아키텍쳐 : ARM, X86_64

▶ Task CPU 및 메모리 for AWS Fargate

256(.25 vCPU)	512MiB, 1GB, 2GB	Linux
512(.5 vCPU)	1GB, 2GB, 3GB, 4GB	Linux
1024(1 vCPU)	2GB, 3GB, 4GB, 5GB, 6GB, 7GB, 8GB	Linux, Windows
2048(2 vCPU)	4~16GB(1GB 증분)	Linux, Windows
4096(4 vCPU)	8~30GB(1GB 증분)	Linux, Windows
8192 (8 vCPU)	16~60GB(4GB 증분)	Linux
16384 (16vCPU)	32~120GB(8GB 증분)	Linux

- 리눅스 컨테이너 리소스 제한 : **ulimits**

[사전준비사항]

1) IAM : 아래 권한을 가진 사용자 IAM 사용자키(csv 파일) 생성후에 CLI 설정 (예시 사용자명 : EcsUser)

AmazonEC2ContainerRegistryFullAccess, AmazonElasticContainerRegistryPublicFullAccess,
AmazonECS_FullAccess, AmazonS3FullAccess(파일다운로드),
IAMFullAccess(역할생성) , CloudWatchFullAccess (Log 그룹생성 삭제)
(익숙치 않을 경우 AdministratorAccess 권한 1 가지를 추가)

2) 작업 EC2 생성 (Name : ECS_Worker) Ubuntu 24.04 / X86-64 / t3.micro / 22 & 80 & 443 Port open 보안그룹
(ssh 접속 / console 혹은 ssh client - ex : putty)

```
sudo apt update -y
sudo apt install zip unzip -y
git clone https://github.com/GaussJung/aws-ecs.git
```

`cd ~/aws-ecs` : docker_example 및 ecs_ref 디렉토리 유무 확인

3) EC2 기동후 CLI 설정 (Linux-x86-64bit)

```
curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscliv2.zip"
unzip -qq awscliv2.zip
sudo ./aws/install
```

(이후 aws configure 진행 – IAM 사용자키 적용)

```
rm -rf aws*
```

4) 도커설정

```
curl -fsSL https://get.docker.com -o get-docker.sh
sudo sh get-docker.sh
sudo apt-get install -y docker-compose
rm get-docker.sh
```

※ rootless 모드 도커활용

```
sudo apt-get install -y uidmap
dockerd-rootless-setuptool.sh install
```

(오류발생 예방을 위해 아래 2 개 진행)

```
sudo setcap cap_net_bind_service=ep $(which rootlesskit)
systemctl --user restart docker
```

5) 샘플 Docker 예제 Registry 등록

`cd ~/aws-ecs/docker_example/cityhome` : 명령이 실행되지 않는다면 git clone 먼저 진행 (위 EC2 생성 참조)

(이미지작성 및 구동)

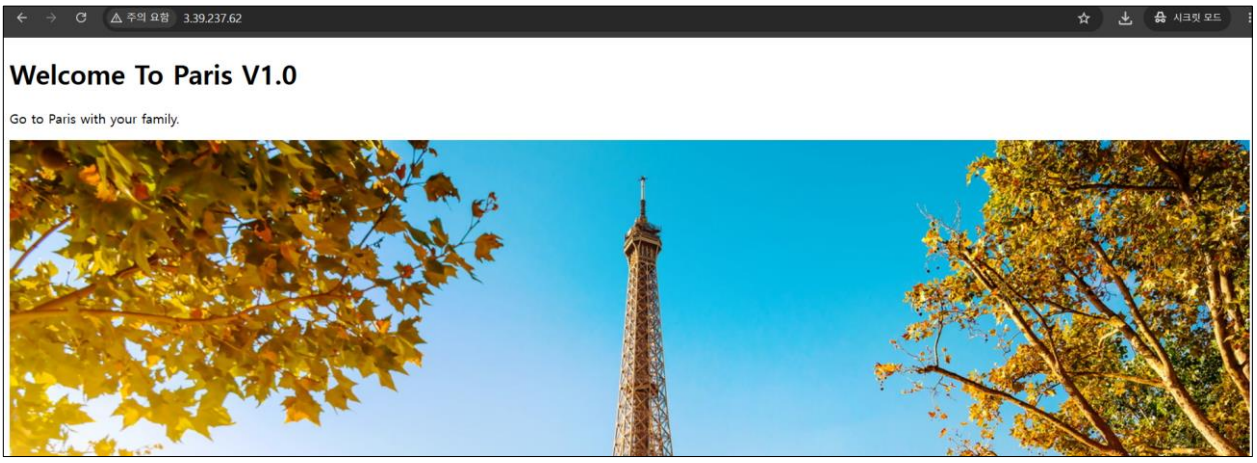
`docker compose up -d --build` : 파라미터 --build (도커이미지생성)

※ 위의 명령은 이미지작성과 컨테이너구동 동시에 진행

```
cd ~/aws-ecs/docker_example/cityhome/webapp_docker
```

```
docker build -t cityhome . && docker run -it -d -p 80:80 --name city80 cityhome
```

웹브라우저에서 퍼블릭 IP 혹은 DNS 접속하여 화면확인



[환경변수]

`export myAwsAccount=111122223333` (※ 비고. 11112222333 은 콘솔우측 상단에 표기된 계정번호로 변경)

[ECR 생성]

```
aws ecr create-repository --repository-name cityhome --region ap-northeast-2
```

[로그인]

```
aws ecr get-login-password --region ap-northeast-2 | docker login --username AWS --password-stdin ${myAwsAccount}.dkr.ecr.ap-northeast-2.amazonaws.com
```

[태깅]

```
docker tag  cityhome:latest  ${myAwsAccount}.dkr.ecr.ap-northeast-2.amazonaws.com/cityhome:latest
```

[푸쉬]

```
docker push ${myAwsAccount}.dkr.ecr.ap-northeast-2.amazonaws.com/cityhome:latest
```

6) IAM 설정

: 역할생성 (IAM 전체권한 혹은 관리자 권한 임시설정)

cd ~/aws-ecs/ecs_ref

```
# ECS Task 실행 역할 생성 :  ECS/Fargate 가 ECR 이미지 Pull, CloudWatch Logs 전송 등에 사용
aws iam create-role  \
  --role-name myrole_ecsTaskExecution  \
  --assume-role-policy-document  file://ecs-tasks-trust-policy.json

# ECS Task 역할 생성 : 컨테이너 내부 애플리케이션이 S3, DynamoDB, SQS 등 AWS API 호출 시 사용
aws iam create-role  \
  --role-name myrole_ecsTaskRole  \
  --assume-role-policy-document  file://ecs-tasks-trust-policy.json
```

========== ecs-tasks-trust-policy.json 내용 ==========

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AllowEcsTasksAssumeRole",
      "Effect": "Allow",
      "Principal": {
        "Service": "ecs-tasks.amazonaws.com"
      },
      "Action": "sts:AssumeRole"
    }
  ]
}
```

▶ myrole_ecsTaskExecution 정책붙임 (관리형정책)

```
aws iam attach-role-policy  \
  --role-name myrole_ecsTaskExecution  \
  --policy-arn arn:aws:iam::aws:policy/service-role/AmazonECSTaskExecutionRolePolicy
```

▶ myrole_ecsTaskRole 정책붙임 (인라인정책)

```
aws iam put-role-policy  \
  --role-name myrole_ecsTaskRole \
  --policy-name mypolicy_ecsTaskAppAccess \
  --policy-document file://mypolicy_ecs_task.json
```

[AWS Fargate 콘솔 : Linux CLI – 사례 WebPage Example (Fargate Spot)]
CLI 참조 : https://docs.aws.amazon.com/cli/latest/reference/ecs/index.html

▶ STEP2-1. 클러스터 생성 :

```
aws ecs create-cluster  --cluster-name  myEcsSpotCluster  --capacity-providers FARGATE
FARGATE_SPOT --region ap-northeast-2
```


```
{
    "cluster": {
        "clusterArn": "arn:aws:ecs:ap-northeast-2:[redacted]:cluster/myEcsSpotCluster",
        "clusterName": "myEcsSpotCluster",
        "status": "ACTIVE",
        "registeredContainerInstancesCount": 0,
        "runningTasksCount": 0,
        "pendingTasksCount": 0,
        "activeServicesCount": 0,
        "statistics": [],
        "tags": [],
        "settings": [
            {
                "name": "containerInsights",
                "value": "disabled"
            }
        ],
        "capacityProviders": [
            "FARGATE",
            "FARGATE_SPOT"
        ],
        "defaultCapacityProviderStrategy": []
    }
}
```


▶ STEP2-2. 테스크 생성

[Log 그룹생성 : 설정파일 myEcsWebServer-spot.json 의 logConfiguration 에서 참조됨]

```
aws logs create-log-group --log-group-name    /ecs/myEcsSpotWebServer
```

[테스크정의]

(비고. myEcsWebServer-spot.json 파일의 11112222333 을 계정번호로 변경할것 - executionRoleArn 과 ECR Image repository 부분)

```
aws ecs register-task-definition  --cli-input-json  file://myEcsWebServer-spot.json
```

위의 작업은 반복시마다 태스크정의가 개정되어짐 (myEcsSpotWebFamily:1 --> myEcsSpotWebFamily:2)

※ 테스크삭제 (잘못만들어졌을 경우)

```
aws ecs deregister-task-definition  --task-definition myEcsSpotWebFamily:1
aws ecs delete-task-definitions   --task-definition myEcsSpotWebFamily:1
```

---------- myEcsWebServer-spot.json ----------

```
 {
 {
   "family": "myEcsWebFamily",
   "networkMode": "awsvpc",
   "containerDefinitions": [
      {
         "name": "myEcsSpotWebServer",
         "image": "111122223333.dkr.ecr.ap-northeast-2.amazonaws.com/cityhome:latest",
         "cpu": 0,
         "portMappings": [
            {
               "containerPort": 80,
               "hostPort": 80,
               "protocol": "tcp"
            }
         ],
         "logConfiguration": {
            "logDriver": "awslogs",
            "options": {
```

```
                "awslogs-group": "/ecs/myEcsSpotWebServer",
                "awslogs-region": "ap-northeast-2",
                "awslogs-create-group": "true",
                "awslogs-stream-prefix": "ecs"
            }
         },
         "healthCheck": {
               "retries": 3,
               "command": [
               "CMD-SHELL",
               "curl -f http://localhost:80/ || exit 1"
            ],
            "timeout": 5,
            "interval": 60,
            "startPeriod": 30
         },
         "essential": true
      }
   ],
   "executionRoleArn": "arn:aws:iam::111122223333:role/myrole_ecsTaskExecutionRole",
   "requiresCompatibilities": [
      "FARGATE"
   ],
   "cpu": "256",
   "memory": "512",
   "runtimePlatform": {
      "cpuArchitecture": "X86_64",
      "operatingSystemFamily": "LINUX"
   },
   "tags": [
     {
        "key": "Name",
        "value": "myEcsSpotWebServer"
     }
  ]
}
------------------------------ EOF myEcsWebServer-spot.json ------------
```

(실행결과 출력 : 생성된 Task 정의 --> 콘솔에서도 확인가능)


```
{
   "taskDefinition": {
      "taskDefinitionArn": "arn:aws:ecs:ap-northeast-2:111122223333:task-definition/myEcsSpotWebFamily:1",
      "containerDefinitions": [
         {
            "name": "myEcsSpotWebServer",
            "image": "111122223333.dkr.ecr.ap-northeast-2.amazonaws.com/cityhome:latest",
            "cpu": 0,
            "portMappings": [
               {
                  "containerPort": 80,
                  "hostPort": 80,
                  "protocol": "tcp"
               }
            ],
            "essential": true,
            "environment": [],
            "mountPoints": [],
            "volumesFrom": [],
            "logConfiguration": {
               "logDriver": "awslogs",
               "options": {
                  "awslogs-group": "/ecs/myEcsSpotWebServer",
                  "awslogs-create-group": "true",
                  "awslogs-region": "ap-northeast-2",
                  "awslogs-stream-prefix": "ecs"
               }
            },
            "healthCheck": {
               "command": [
                  "CMD-SHELL",
                  "curl -f http://localhost:80/ || exit 1"
               ],
               "interval": 60,
               "timeout": 5,
               "retries": 3,
               "startPeriod": 30
            },
            "systemControls": []
         }
      ],
      "family": "myEcsSpotWebFamily",
      "executionRoleArn": "arn:aws:iam::111122223333:role/ myrole_ecsTaskExecutionRole",
      "networkMode": "awsvpc",
      "revision": 5,
      "volumes": [],
      "status": "ACTIVE",
      "requiresAttributes": [
         {
            "name": "com.amazonaws.ecs.capability.logging-driver.awslogs"
         },
         {
            "name": "ecs.capability.execution-role-awslogs"
         },
         {
            "name": "com.amazonaws.ecs.capability.ecr-auth"
         },
         {
            "name": "com.amazonaws.ecs.capability.docker-remote-api.1.19"
         },
         {
            "name": "ecs.capability.container-health-check"
         },
         {
            "name": "ecs.capability.execution-role-ecr-pull"
         },
         {
            "name": "com.amazonaws.ecs.capability.docker-remote-api.1.18"
         },
         {
            "name": "ecs.capability.task-eni"
         },
         {
            "name": "com.amazonaws.ecs.capability.docker-remote-api.1.29"
         }
      ],
      "placementConstraints": [],
      "compatibilities": [
         "EC2",
         "FARGATE"
      ],
      "runtimePlatform": {
         "cpuArchitecture": "X86_64",
```

```
            "operatingSystemFamily": "LINUX"
        },
        "requiresCompatibilities": [
            "FARGATE"
        ],
        "cpu": "256",
        "memory": "512",
        "registeredAt": "20XX-XX-04T20:13:03.018000+09:00",
        "registeredBy": "arn:aws:iam::111122223333:user/containeruser"
    },
    "tags": [
        {
            "key": "Name",
            "value": "myEcsSpotWebServer"
        }
    ]
}
```

▶ STEP2-3. 서비스 생성

https://docs.aws.amazon.com/AmazonECS/latest/APIReference/API_CreateService.html

확인사항 : subnet-ID, 보안그룹-ID

```
aws ecs create-service \
  --cluster myEcsSpotCluster \
  --service-name myEcsSpotService \
  --network-configuration "awsvpcConfiguration={subnets=[subnet-bbaa91111,subnet-
ddcc2222],securityGroups=[sg-aaaa1111bbbb],assignPublicIp=ENABLED}" \
  --cli-input-json file://myEcsSpotService.json
```

```
------------ myEcsSpotService.json ------------
{
   "taskDefinition": "myEcsSpotWebFamily:1",
   "desiredCount": 2,
   "capacityProviderStrategy": [
      {
         "base": 0,
         "capacityProvider": "FARGATE_SPOT",
         "weight": 1
      }
   ],
   "tags": [
      {
         "key":"Name",
         "value":"myEcsSpotService"
      },
      {
         "key":"Sector",
         "value":"MyTraining"
      }
   ]
}
```

ECS Service 에서는 Capacity Provider Strategy 를 사용하여 일반 Fargate 와 FARGATE_SPOT 을 혼합 운영할 수 있음
base 값은 최소 유지할 Task 수량을 의미하며, weight 값은 추가 Task 를 어떤 비율로 분산 배치할지를 의미한다.

예를 들어 FARGATE 의 weight=1, FARGATE_SPOT 의 weight=3 으로 설정하면 추가 Task 는 약 1:3 비율로 배치됨.
이를 통해 안정성을 유지하면서도 운영 비용을 절감할 수 있음.

```
------------ myEcsMixedService.json ------------
{
  "taskDefinition": "myEcsMixedFamily",
  "desiredCount": 2,
  "capacityProviderStrategy": [
    {
      "capacityProvider": "FARGATE",
      "base": 1,
      "weight": 1
    },
    {
      "capacityProvider": "FARGATE_SPOT",
      "weight": 3
    }
  ],
  "tags": [
    {
      "key": "Name",
      "value": "myEcsMixedService"
    },
    {
      "key": "Sector",
      "value": "Production"
    }
  ]
}
```

비고) 서비스 삭제 (테스트수량조절 후에 삭제)

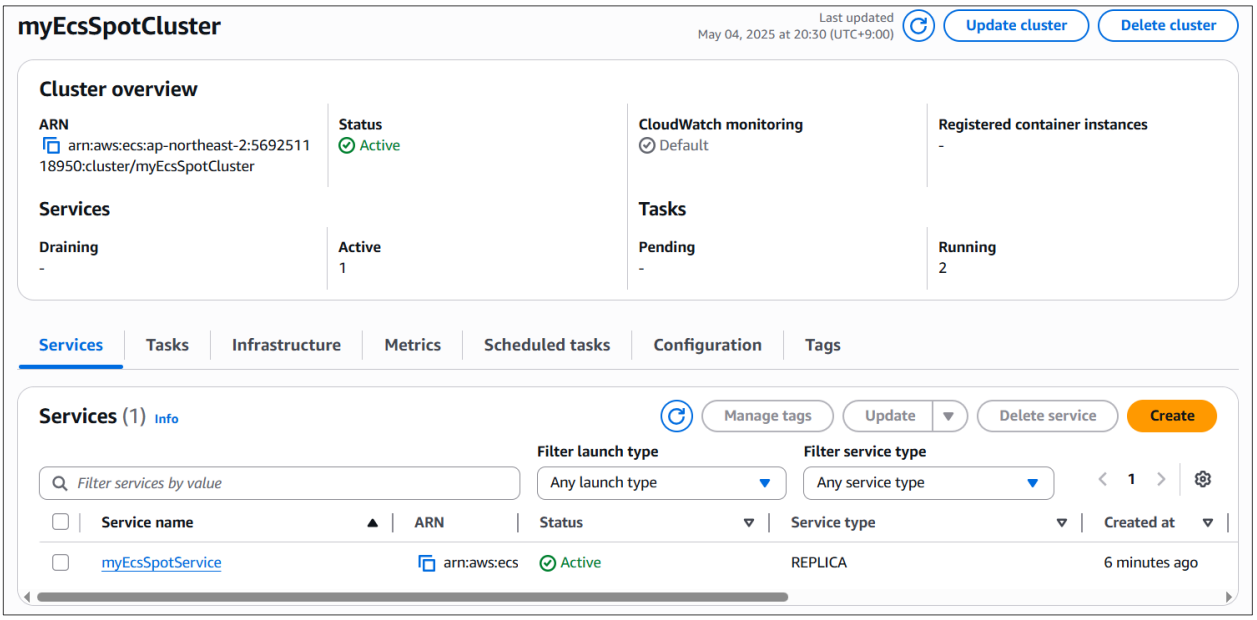
```
aws ecs update-service  --cluster myEcsSpotCluster --service myEcsSpotService --desired-count 0

aws ecs delete-service  --cluster myEcsSpotCluster --service myEcsSpotService
```

도커이미지를 새롭게 작성하여 Push 하였다면 Task 를 다시 등록하고 서비스 업데이트 진행

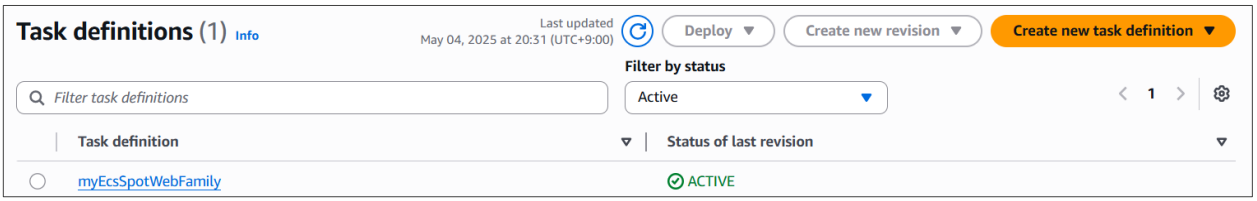
```
aws ecs register-task-definition
 --cli-input-json  file://myEcsWebServer-spot.json

aws ecs update-service \
  --cluster myEcsSpotCluster \
  --service myEcsSpotService \
  --task-definition myEcsWebFamily
```

[콘솔에서 생성확인]

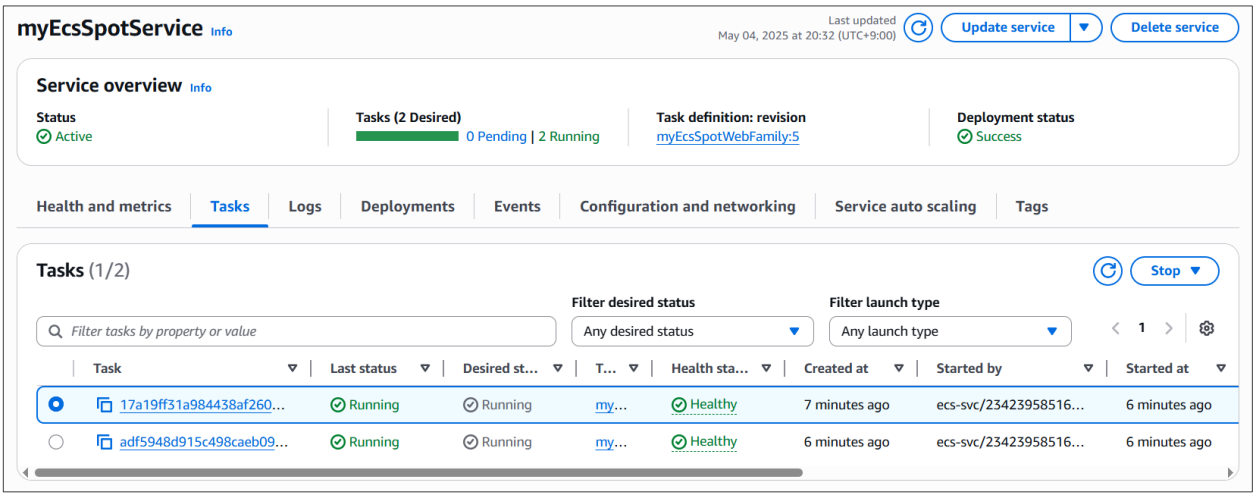
(클러스터)



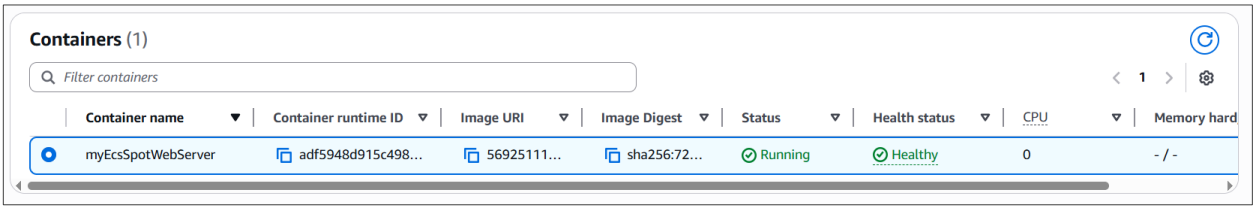
(테스크정의)



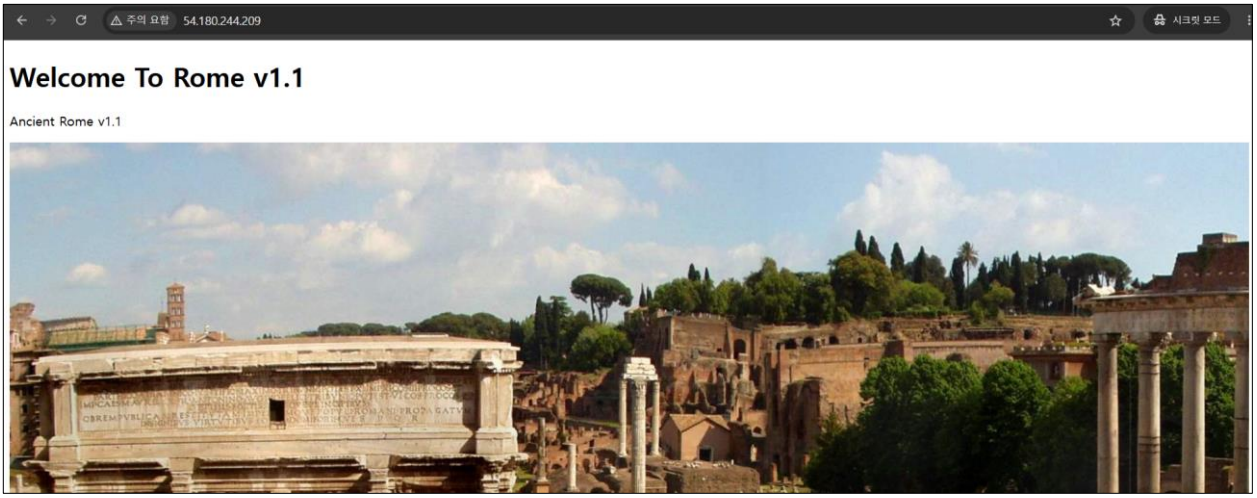
(서비스)



(Task 하단 컨테이너 정의)



(Task 접속)
Task 에 할당된 Public IP 로 웹브라우져에 붙이고 접속 테스트



▶ 로드밸런서 + 서비스 AutoScaling 적용

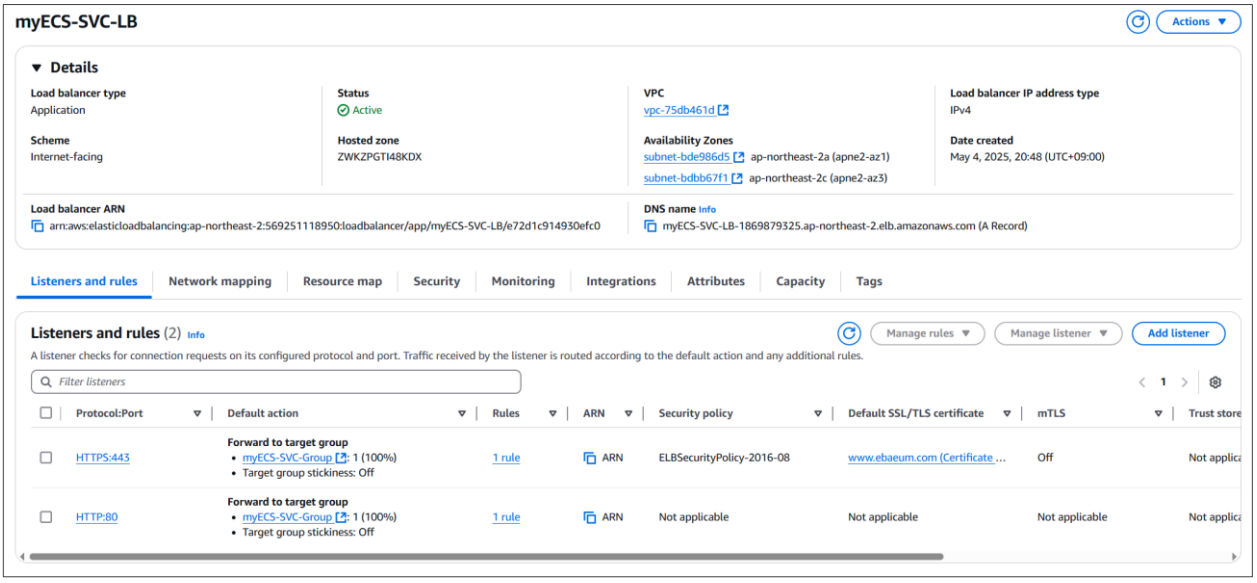
사전작업 - 자원생성
1. 로드밸런서 생성
```
aws elbv2 create-load-balancer --name myECS-SVC-LB --subnets subnet-bde91111 subnet-bdbb2222 --security-groups sg-045490d9f04b67e3f --region ap-northeast-2
```

2. 목표그룹 생성
```
aws elbv2 create-target-group --name myECS-SVC-Group --target-type ip --protocol HTTP --port 80 --vpc-id vpc-75db461d --ip-address-type ipv4 --region ap-northeast-2
```

3. 리스너 생성

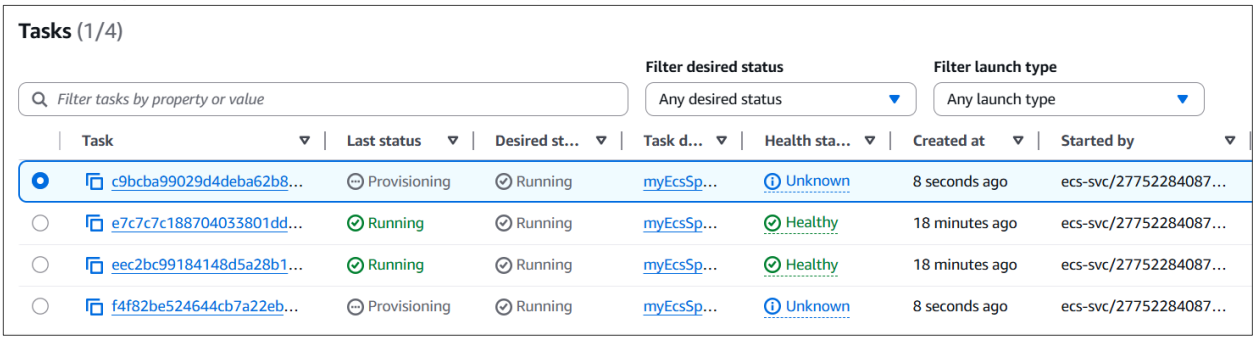
(80) 아래 로드밸런서 및 목표그룹 ARN 을 콘솔에서 확인하여 변경함. (붉은색상)
```
aws elbv2 create-listener --load-balancer-arn arn:aws:elasticloadbalancing:ap-northeast-2:111122223333:loadbalancer/app/myECS-SVC-LB/e72d1c914930efc0 --protocol HTTP --port 80 --default-actions Type=forward,TargetGroupArn=arn:aws:elasticloadbalancing:ap-northeast-2:111122223333:targetgroup/myECS-SVC-Group/b8be659e53280785
```

(443) 아래 인증서, 로드밸런서 및 목표그룹 ARN 을 콘솔에서 확인하여 변경함. (붉은색상)
```
aws elbv2 create-listener --load-balancer-arn arn:aws:elasticloadbalancing:ap-northeast-2:111122223333:loadbalancer/app/myECS-SVC-LB/e72d1c914930efc0 --protocol HTTPS --port 443 --certificates CertificateArn=arn:aws:acm:ap-northeast-2:111122223333:certificate/63ec49f5-717e-448f-89d1-49cfd95c9e3f --default-actions Type=forward,TargetGroupArn=arn:aws:elasticloadbalancing:ap-northeast-2:111122223333:targetgroup/myECS-SVC-Group/b8be659e53280785
```



(※ Public IP 할당해제 --> 이 경우 NAT Gateway 혹은 VPC EndPoint 가 설정되어 있어야 Task 동작함)
```
aws ecs update-service --cluster myEcsSpotCluster --service myEcsSpotService --network-configuration "awsvpcConfiguration={subnets=[' subnet-bde986d5 ','subnet-bdbb67f1'],securityGroups=[sg-045490d9f04b67e3f], assignPublicIp=DISABLED}"
```

로드밸런서 적용
(참조 : https://docs.aws.amazon.com/cli/latest/reference/ecs/update-service.html)
[Task 수량 조절]
(4 개로 조절)
```
aws ecs update-service  --cluster myEcsSpotCluster --service myEcsSpotService --desired-count 4
```



(0 개로 조절 : 즉, Task 작동 중지)
```
aws ecs update-service  --cluster myEcsSpotCluster --service myEcsSpotService --desired-count 0
```

[서비스에 로드밸런서 적용 + Task 수량 3 개로 조정]
```
aws ecs update-service --cluster myEcsSpotCluster --service myEcsSpotService --desired-count 3
--load-balancers  "targetGroupArn=arn:aws:elasticloadbalancing:ap-northeast-
2:111122223333:targetgroup/myECS-SVC-
Group/b8be659e53280785,containerName=myEcsSpotWebServer,containerPort=80"
```

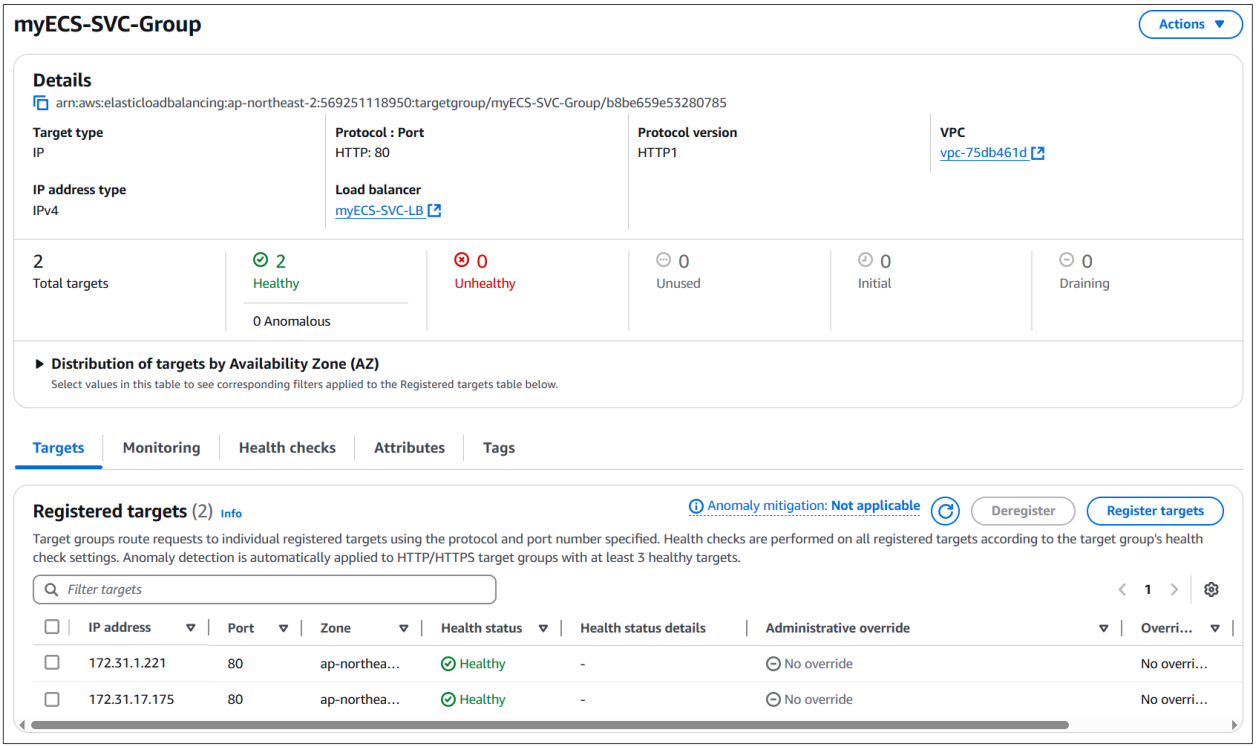
[서비스에 로드밸런서 적용 (수량조절없이)]
```
aws ecs update-service --cluster myEcsSpotCluster --service myEcsSpotService  --load-balancers
"targetGroupArn=arn:aws:elasticloadbalancing:ap-northeast-2:111122223333:targetgroup/myECS-SVC-
Group/b8be659e53280785,containerName=myEcsSpotWebServer,containerPort=80"
```

[로드밸런서 제거]
```
aws ecs update-service \
  --cluster myEcsSpotCluster \
  --service myEcsSpotService \
  --load-balancers '[]'
```

[로드밸런서 PublicDNS 혹은 인증서 설정 SSL URL 로 접속함]

--> Task 로 접속했을 때와 동일한 화면 출력

[목표그룹에 등록된 ECS 서비스 Task]



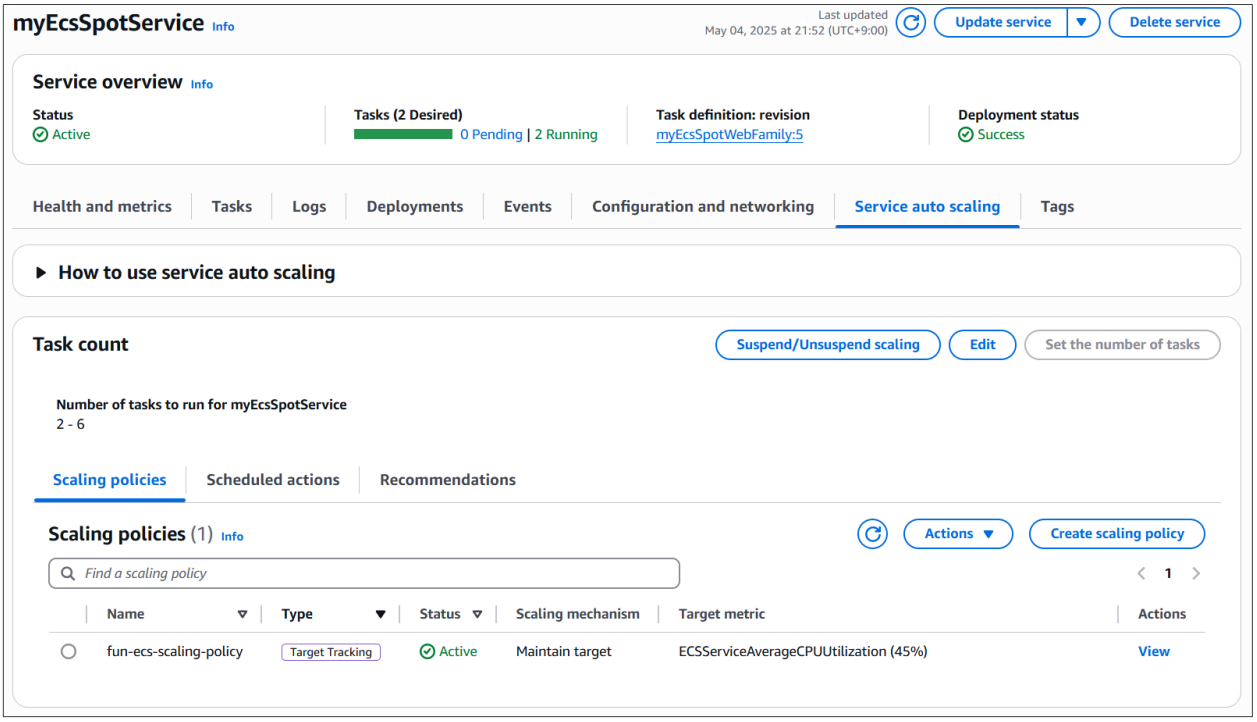
▶ 서비스 오토스케일링 정책 적용

참조) https://repost.aws/knowledge-center/ecs-fargate-service-auto-scaling

[오토스케일링]

```
- (선택) Task Definition 이 업데이트된 경우 서비스에 최신 Revision 반영
aws ecs update-service \
  --cluster myEcsSpotCluster \
  --service myEcsSpotService \
  --task-definition myEcsWebFamily


- 사전수행 0 개로 변경 (필요시 --> 단, 권장사항 아님)

aws ecs update-service  \
--cluster myEcsSpotCluster \
--service myEcsSpotService  \
--desired-count 0


A1) 어플리케이션 오토스케일링 타겟 설정 ( 2 개 ~ 6 개)

- 중요사항 : Resource-ID : 클러스터명 + 서비스명

aws application-autoscaling register-scalable-target \

--service-namespace ecs --scalable-dimension ecs:service:DesiredCount  \

--resource-id service/myEcsSpotCluster/myEcsSpotService \

--min-capacity 2 --max-capacity 6 --region ap-northeast-2

( 결과 예시)
{
    "ScalableTargetARN": "arn:aws:application-autoscaling:ap-northeast-2:111122223333:scalable-
target/0ec5de7fd7704c4548278e1e60d617278376"
}


A2) 추적정책등록-CPU (CPU 이용률 50% 초과시 자원증대, 확장시 70 초간 확인, 수축시 60 초 확인)
aws application-autoscaling put-scaling-policy \
  --service-namespace ecs \
  --scalable-dimension ecs:service:DesiredCount \
  --resource-id service/myEcsSpotCluster/myEcsSpotService \
  --policy-name my-ecs-scaling-policy \
  --policy-type TargetTrackingScaling \
  --target-tracking-scaling-policy-configuration
'{"TargetValue":50.0,"PredefinedMetricSpecification":{"PredefinedMetricType":"ECSServiceAverageC
PUUtilization"},"ScaleOutCooldown":70,"ScaleInCooldown":60}' \
  --region ap-northeast-2

(결과예시)
{
    "PolicyARN": "arn:aws:autoscaling:ap-northeast-2:111122223333:scalingPolicy:de7fd770-4c45-4827-8e1e-
60d617278376:resource/ecs/service/myEcsSpotCluster/myEcsSpotService:policyName/my-ecs-scaling-policy",
    "Alarms": [
        {
            "AlarmName": "TargetTracking-service/myEcsSpotCluster/myEcsSpotService-AlarmHigh-7ed79954-173f-435e-
9597-98965851f7ff",
            "AlarmARN": "arn:aws:cloudwatch:ap-northeast-2:111122223333:alarm:TargetTracking-
service/myEcsSpotCluster/myEcsSpotService-AlarmHigh-7ed79954-173f-435e-9597-98965851f7ff"
        },
        {
            "AlarmName": "TargetTracking-service/myEcsSpotCluster/myEcsSpotService-AlarmLow-a17e9386-2a58-4ca3-
b9ed-459821a36aca",
            "AlarmARN": "arn:aws:cloudwatch:ap-northeast-2:111122223333:alarm:TargetTracking-
service/myEcsSpotCluster/myEcsSpotService-AlarmLow-a17e9386-2a58-4ca3-b9ed-459821a36aca"
        }
    ]
}
```

[콘솔확인]



[자원삭제]

```
: ECS Task 및 서비스삭제 (콘솔 삭제 혹은 아래 CLI)
aws ecs update-service  --cluster myEcsSpotCluster  --service myEcsSpotService --desired-count 0
aws ecs delete-service   --cluster myEcsSpotCluster   --service myEcsSpotService

: LoadBalancer 삭제 (콘솔 삭제 혹은 아래 CLI)
aws elbv2 delete-load-balancer --load-balancer-arn arn:aws:elasticloadbalancing:ap-northeast-
2:111122223333:loadbalancer/app/myECS-SVC-LB/7253f29426aec20b

: Target Group 삭제  (콘솔 삭제 혹은 아래 CLI)
aws elbv2 delete-target-group --target-group-arn arn:aws:elasticloadbalancing:ap-northeast-
2:111122223333:targetgroup/myECS-SVC-Group/e500d2afeea00c58

: 오토스캐일링 정책제거
aws application-autoscaling deregister-scalable-target \
  --service-namespace ecs \
  --scalable-dimension ecs:service:DesiredCount \
  --resource-id service/myEcsSpotCluster/myEcsSpotService \
  --region ap-northeast-2

: ECS Task 등록해제(수량만큼)
aws ecs deregister-task-definition --task-definition myEcsWebFamily:1
aws ecs deregister-task-definition --task-definition myEcsWebFamily:2
aws ecs deregister-task-definition --task-definition myEcsWebFamily:3

: ECS Task 삭제 (수량만큼)
aws ecs delete-task-definitions \
  --task-definitions \
  myEcsWebFamily:1 \
  myEcsWebFamily:2 \
  myEcsWebFamily:3 \

: ECS Cluster 삭제
aws ecs delete-cluster --cluster myEcsSpotCluster

: Log 그룹삭제
aws logs delete-log-group --log-group-name   /ecs/myEcsSpotWebServer
```

– 끝 –